CSC 522 Project Proposal

CSC 522 Project Proposal – Design Document

Instructions

- If you haven't already done so, please complete the pre-survey.
- Before you start, read over the project requirements (posted on Moodle), in particular the novelty requirements and dataset suggestions. Find a dataset and brainstorm some project ideas with your team. When you're ready, start this document.
- You do not have to work in order, and you can split up primary responsibility for different sections. Make sure everyone in your group is able to justify all major design decisions.
- If you see a text box, enter your answer in the box.
- Pay attention to the highlighted instructions indicating when to complete/skip certain sections (depending on your dataset and project goals).
    - If a section does not apply to your project, put N/A as your answer.
- If you see a multiple-choice question, select one option (or all that apply), e.g., by highlighting it or **bold underlining it**.
- If you choose "Other (describe)", describe your answer in the text box below.
- Many questions ask you to **explain** or **justify** your decision. Make sure to do so.
    - If you are *unsure* how to justify your answer or whether you've made the best decision, you can say that (don't just make something up!).
- Most of these questions should be familiar from the lectures and guided projects, but a few may be new to you. If you are unsure how to proceed after reading the instructions carefully, post a question on Ed.

Note: If the project you want to propose does not fit well into this design document structure, you should plan to **meet with course staff** in office hours or on Ed before proceeding to make sure it is well-scoped. We can discuss how to modify your proposal document.

Also note that there are **3 sections** to this design document, and you should fill out at least 2 of them:

- The main section here
- The classification section is applicable only if you are doing classification.
- The regression section is applicable only if you are doing regression.

1. Problem Description

These questions ask you to describe your project and dataset at a high level.

1. **List all your team members (First name, Last name, UnityID):**

William Hansen (whansen)
Brian Marks (bdmarks)
Ryan McPhee (rcmcphee)

2. **Abstract**: Give a 1-paragraph overview of your project, including the problem you are trying to solve, why it matters, and how you will attempt to solve it.

The scope of the internet seemingly grows every minute, with more information, resources, and data being continuously integrated into every possible fold. However, a significant proportion of internet nodes exude undesirable features, poised to disrupt or infect victims for the benefit of potential adversaries through methods such as spam, phishing, malware, and defacement. In this paper, we develop a learning model trained on an established database of malicious URL details of various calibers to attempt to accurately predict if a presented URL is malicious or not and what variant of malicious activity it may present with hopes of increasing cybersecurity measures through better tuning of alert and detection policies of email rules, web filtering, and network intrusion detection systems.

3. **Share link(s) to the dataset(s) you will use. If not publicly accessible, or if you will create your own dataset, briefly describe how to obtain it (source, provider, etc).** Example: "https://www.kaggle.com/datasets/... or description of how to obtain.

https://www.unb.ca/cic/datasets/url-2016.html

4. **Do you plan to link or combine your multiple data sources, or are they used for separate experiments? If you plan to combine datasets, record the steps you took (or will take) to create the final dataset so someone else can reproduce it.** Include how your datasets are linked. If they have different units of analysis, explain how they will be unified. Record the steps you took (or will take) to create the final dataset so someone else can reproduce it. Example: “The HR table is the primary table; the performance table adds quarterly metrics. Steps: 1. Load HR data... 2. Join keys: employee_id. Join type: left join."

We will use a single dataset.

5. **Prediction task(s) and target column name(s).** Write one sentence per prediction task. Use this format: “Given [inputs], predict [target] for [unit of analysis] (target column: COLUMN_NAME).” If you have multiple tasks, put each on a new line.

Example: Given employee attributes, predict whether the employee will leave the company for each employee (target column: Attrition).

Given the lexical characteristics of a URL, predict whether the content is spam, malware, phishing, defacement, or benign.

6. **What type of supervised learning problem is this?**
   - Classification
   - Regression

7. **Research Questions (RQs)**

List 2–4 research questions (RQs) your project will answer. For each RQ, briefly explain how you will answer it (what section, metric, or analysis you will use).

1. **Which attributes are most meaningful when predicting whether a URL is malicious or not?**
   a. Run several trials, each independently, with the same trained model, with one attribute of the dataset left out. Compare the overall accuracy and precision of the prediction results to see which is the lowest and therefore the most influential attribute in correct predictions.
2. **Which attributes are most meaningful when predicting whether a malicious URL is spam, phishing, malware, or defacement?**
   a. Run several trials, each independently, with the same trained model, with one attribute of the dataset left out. Compare the overall accuracy and precision of the prediction results to see which is the lowest and therefore the most influential attribute in correct predictions.

Examples (not exhaustive):

- **Model performance:** Which model performs best on the primary metric? All projects will likely include the RQ.
- **Preprocessing impact:** Does a preprocessing choice (e.g., scaling, imputation strategy, discretization, sampling/SMOTE) improve performance?
- **Feature importance:** What attributes meaningfully predict the target?
- **Interpretability-Performance tradeoffs:** How well does a simple, human-interpretable model (e.g., shallow decision tree, regression, etc.) perform compared to a more complex model?
- **Fairness/bias (if applicable):** Do predictions differ across protected groups? Can/should these differences be mitigated?

8. **Novel Aspects**: Describe at least two novel aspects of your project, which go beyond what we have done in the guided projects. See the project description on Moodle for examples of novel aspects. These may be related to your RQs above.

Novel Aspect 1: Feature Engineering

Oftentimes, threat actors will register and use potential misspellings (microsofft.com or micrrosoft.com) of well-known websites and URLs to capture typos by users, as well as disguised URLs (rnicrosoft.com or nnicrosoft.com) to mimic well-known websites and trick users. We will aim to incorporate some of this knowledge and logic into our model(s) to see if they have any impact on predictive results and accuracy.

Novel Aspect 2: Automated Tuning

Rather than tuning the model's hyperparameters manually, we will try to use a machine learning technique like Bayesian optimization to tune them more quickly and precisely.

2. Data

The following subsections all concern your data and how you will process it.

2.1. Non-tabular Data

This section asks you to describe how you will process non-tabular data (e.g., audio, images), if applicable.

9. **Identify the format of your final dataset (after pre-processing) that will be used for training.** (Select all that apply):

- Tabular (rows and columns)
- Non-tabular (e.g., images, audio, time-series, etc. – describe below)
- Other

If "Other": Specify other type:

N/A

Answer the next two questions if applicable; if not, fill "N/A."

10. **How will you encode your non-tabular data?**

- Use a model designed for this data type directly (e.g., deep learning on images/text/audio)
- Extract tabular features from each instance, then train a model on tabular data

- Other (describe)

N/A

11. **Describe what numbers (features) you will compute from each data example before training your model. If you compute many similar features, you can describe them collectively.**

N/A

What to include:

- What you'll compute (feature names or a short description)
- How you'll compute it (tool/method, and any key settings like a time window)
- About how many features you'll end up with (rough estimate is fine)

Example: Group 1: Time-series summary stats
I will compute mean, min, max, and standard deviation for each sensor channel using 30-second windows. This gives about 4 stats × 12 channels = 48 features.

2.2. Data Validation

These questions will walk you through best practices for validating and preprocessing your dataset.

12. **Do any of your features have inconsistent datatypes (numeric vs categorical vs date/time) or were they parsed incorrectly?** For example, a column may be clearly numeric but include string values like "N/A" or "None", which cause Pandas to parse the column as a string.
    - Yes
    - No

If "**Yes**": Briefly list any corrections you need to make ("Column X should be categorical, not numeric").

Some fields use -1 as a placeholder to distinguish something as not present. The data also has some true missing values with "NaN" in several numeric columns.

13. **Do any feature(s) directly or indirectly reveal the target attribute (columns derived from the target, post-outcome information)?**

- Yes
- No

If "**Yes**": List the column(s) and explain why they may leak target information. What would you do to address it?

N/A

14. **Did you find outliers/extreme values in any features?**
    - Yes
    - No

If "Yes": **What criteria did you use to define an outlier?**

- Visual inspection using histograms
- Visual inspection using box plots
- Z-score threshold (e.g., |z| > 3)
- Other (describe)
- N/A

N/A

15. **Handling extreme values:** For the extreme values (outliers), are they useful (or potentially useful) for predicting your target variable, or are they not useful because they are likely noise/errors (e.g., data entry mistakes, measurement glitches, or otherwise unreliable values)?
    Tip: extreme values can throw off preprocessing steps like normalization and certain models, but they can also carry useful information (e.g., maybe they are more likely to have a given class).
    - Mostly meaningful (keep)
    - Mostly noise/errors (remove or cap)
    - Mixed (case-by-case)
    - I'm not sure
    - Other (describe)
    - N/A

Describe your approach, if any, for handling extreme values:

The extreme values look mostly meaningful rather than like bad data. Many features are counts, lengths, and ratios, so they are naturally skewed and heavy-tailed. We will not remove these values by default. Instead, we will use robust preprocessing, and for linear models, we may apply log-style transforms or standardization to reduce skew.

Reminder: Common approaches include removal, capping/clipping, discretization/binning, transformations (e.g., log), robust scaling, or using models less sensitive to outliers.

2.3. Data Groups

These questions ask about potentially meaningful groups in your dataset and how you will handle them.

16. **How is one instance represented in the dataset?**

Reminder: we make one prediction per instance. This often corresponds to one row in a dataset; however, sometimes a dataset has multiple rows per instance, e.g., multiple readings of a patient's vital signs over time, but we just make one prediction per patient.

- One row per instance/prediction
- Multiple rows per instance/prediction
- One file per instance/prediction
- Other.

If *other*:

N/A

If “One row per instance/prediction”, skip the rest of this section.

17. **How will you aggregate multiple rows into one instance per group for modeling?**
    Examples: mean/max over time, last observation, counts, rolling/window summary, one row per period, etc.

N/A

18. **Are there duplicate or redundant records that could affect grouping/aggregation (e.g., the same entity recorded more than once for the same time period/event)?**

- No
- Yes
- Not applicable

If “Yes”: **How will you determine whether duplicates are meaningful vs. erroneous, and what will you do about them?**

We will check for exact duplicate rows and remove them before splitting to prevent data leakage. Duplicate rows do not represent repeated real events.

Example:

- Will you check for exact duplicate rows and/or duplicate IDs?
- Do duplicates represent repeated real events (meaningful) or accidental duplication (erroneous)?
- If duplicate IDs have conflicting values, how will you resolve conflicts?
- If you aggregate (e.g., per user/month), will you deduplicate before aggregation or combine duplicates during aggregation?

2.4. Data Splitting

19. **What approach will you use for the final evaluation of your models (test evaluation)?** *Reminder: we're not asking about hyperparameter tuning; this is for the **final** evaluation.*
    - Holdout test set (reserve a test split and evaluate once at the end)
    - Cross-validation for final evaluation (evaluate using CV across the dataset)
    - Other (describe)

Justify your answer, and also specify:

- *If using holdout, what percent of data will be used for training vs test datasets?* Example: "10% train, 80% test".
- *If using cross-validation: how many folds will you use? Why is cross-validation necessary?* Examples: small dataset, high variance in performance, need a more robust estimate.

We will use a stratified holdout test set, with about 80% for training and 20% for testing. This is simple, reproducible, and appropriate because the dataset is large enough for a stable split.

20. **Stratification:** Will you keep the target classes in the same proportions in your train/test split (i.e., stratify by the target label)? Why or why not? Briefly justify your choice (e.g., small dataset with a minority class → stratify to keep class balance).
    - Yes
    - No
    - N/A (not a classification problem)

Justify your answer:

We will use stratification so the five classes stay in similar proportions in train and test, especially because ***malware*** is the smallest class. There are four undesirable target values (malware, spam, defacement, and phishing) and one desirable value (benign). Stratification

will ensure that each of these 5 target classes is appropriately represented in our training and test split.

Example: “Yes, because the minority class is small and we want similar class proportions in train and test.”

21. **Test dataset influence**: How will you avoid the test dataset(s) influencing your decisions as you work (especially if using cross-validation, when all data will be part of the test set)?

    Examples: keep model/pipeline decisions fixed before final CV, use nested CV, or use a separate validation strategy for tuning, and reserve CV only for final reporting.

Since we’ll be using a holdout test set, we will set the test data aside before doing any further work and use it only for the final evaluation.

22. **Are there meaningful groups or repeated measurements in your dataset?**

Examples: multiple instances per entity (user/student/patient/account/device/stock), students grouped into different semesters, repeated measurements of the same variable over time, multiple events per customer, etc. You might have meaningful groups even if every row is a distinct instance.

- Yes (remember this for the next question)
- No

If “**Yes**”: **What defines a “group” and/or time ordering in your data?**
Examples: group key (student_id, account_id, stock_ticker), semester, timestamp/date column

N/A

23. **Dependencies and data splitting:** Does your dataset have instances that are related to or dependent on one another? If so, we need to ensure they are properly divided into train/validation/test sets to avoid leaking information. Select all that apply:
    - Multiple measurements over time (past instances should predict future instances)
    - Multiple instances that share a distinct grouping ID (e.g., predicting success for the same student in different classes)

- ○ Other (specify below)

If so, describe how you will split the data between training/validation/test sets so that you don't leak information. Make sure your test data has the same properties that you would expect *unseen* data to have (e.g., test time > training time; test IDs distinct from training IDs). Include the feature name you will use to stratify data, if applicable. If this does not apply, write N/A.

N/A

2.5. Missing Data

This section asks you how you will handle missing data in your dataset (i.e, imputation).
Skip this section if you have only non-tabular data (i.e., no existing or extracted features).

24. **Are there missing values?**
    - No
    - Yes

If "Yes": What missing value strategy are you going to use to handle missing **numeric values**?

- Mean
- Median
- Mode
- Drop rows with missing values
- Model-specific missing handling
- Not applicable
- Other (describe)

We will use median imputation for numeric features rather than dropping columns, to preserve information and avoid unnecessary data loss.

If "Yes": What missing value strategy are you going to use to handle missing **categorical values**

- Most frequent
- Add "Unknown/Missing" category
- Drop rows with missing values
- Not applicable
- Other (describe)

N/A

3. Features: Transformation, Selection, and Engineering

This section asks you to describe how you will transform and select features.
Skip this section if you have only non-tabular data (i.e., no existing or extracted features).

25. **Will you scale/normalize numeric features?**
    - None
    - Standardization (z-score)
    - Min-max Normalization
    - Other (describe).

Justify your answer (Explain how it relates to your model selection.):

Our dataset has rates or attributes on a 0-1 scale and counts that reach the hundreds and even thousands. Because we intend to use a k-NN model, we must scale the features. Min-max normalization might work for this purpose, but z-score standardization is more tolerant of outliers and imbalanced attributes, which aren't uncommon in our dataset.

26. **Functional Transformations:** Will you apply any functional transformations to features (e.g., log transform for skewed numeric features)?

Example:

- Transformations are most helpful when using linear or distance-based models (e.g., Logistic Regression, Linear SVM, k-NN), especially when numeric features are **highly skewed**. To detect skew, inspect histograms/box plots or summary stats (e.g., long right tail).
- income: log1p(income) (reduce right-skew), area: sqrt(area), Note if you standardize after transforming

Justify your answer (**List the feature(s), transformation(s), and any settings):**

Yes, we predict that we will perform functional transformations on the features. Many of our features are skewed toward one of the tails, which would impact the effectiveness of our models of choice. The histogram for things like ***URL_Letter_Count*** have a long right tail, as most URLs are between 0-100 characters, but many get into the thousands, especially with spam URLs.

27. **Will you use feature selection?** (No/Yes/I will compare both (with vs. without feature selection) and choose the best). **How will you apply feature selection (if at all)? List the feature selection method(s) you will use and the starting settings. If you are comparing options, list multiple methods.** Hint: you can include a 'no feature selection' baseline.

Justify your answer **:**

Yes, we will use feature selection. We will begin with a variance threshold to remove attributes that remain static, or near-static, across the data, such as ***isPortEighty***. We will use backward elimination to remove features that have little bearing on the target class and only contribute to noise and computational time.

28. **Are there any features you might engineer that could help with prediction? Briefly describe the feature idea(s) and how you would create them.**

Our starting dataset has an impressive level of detail, including many creative features, such as entropy, ratios, and functions of other features.

Examples:

- Combining existing features, using expert knowledge to define a new feature, aggregations within a group (per user/month), or merging in a column from another dataset.
- New feature name (or idea). Base column(s)/source. Rule/logic (or join/merge description if from another dataset). Why do you think it may help with prediction?

4. Preprocessing Pipeline and Experiments

29. **Preprocessing Pipeline**: Based on your answers above, describe your core preprocessing pipeline in the order it will occur. If you will use different preprocessing pipelines (or variants) for different models, specify that below. Consider including:
    - Encoding approach (for categorical/text)
    - Imputation (missing-value handling)
    - Scaling/normalization

- Functional transformation (e.g., log transform)
- Discretization/binning
- Feature selection/dimensionality reduction
- Sampling (over/undersampling/synthetic) – see the "Classification" tab for more on this
- Other (specify)

Example: Imputation [Attr1, Attr2] → Discretization [Attr1, Attr2] → Encoding → Min-max Normalization → SMOTE

Duplicate Removal → Missing Value Handling (Imputation + -1 cleanup) → Encoding (tld if used) → Functional Transformation (log for skewed features) → Scaling → Feature Selection → Sampling This pipeline first cleans and standardizes the data, then applies transformations and feature reduction, and optionally adjusts class balance, ensuring the model trains on consistent, well-structured inputs while avoiding data leakage.

30. **Experiments**: Will you run experiments to decide whether/how to include any of these steps in your pipelines? If so, for which steps? If you were unsure about any design decisions in this document, this is a good way to address that uncertainty.
Note: We will cover preprocessing *hyperparameters* (e.g., how many features to select) later; this is for including/excluding steps.

Yes, we will run experiments to decide which preprocessing steps help. In particular, we will test whether our feature selection improves performance and how our preprocessing hyperparameters (such as k) affect our predictions.

31. **How will you run the experiment?** Explain:
    - What is your validation/evaluation strategy (e.g., holdout or CV) and metric?
    - Which model(s) will you use to make the decision, or will you make different decisions for different models?
    - If you have multiple experiments, how will you structure them (e.g., try A, then B; try all combinations)?

We will be using holdout evaluation, and our main metric will be the F1 score. We will prioritize the k-NN model. We will compare configurations using controlled experiments rather than sequential manual changes

5. Models

This section asks you to define the models you will compare.

32. **What baseline will you use to establish a minimum bar? (Example:** Majority baseline, Other). Describe your baseline:

Weighted random guessing: a class is selected at random, with a probability based on the prevalence of that class in the training data.

33. **Choose the models you want to compare. For every model, provide a short justification.** Examples: DummyClassifier, Logistic Regression, Naive Bayes (Gaussian/Multinomial), Linear SVM, SVM (RBF kernel), Ridge/Lasso (for regression tasks)**,** k-NN, Decision Tree, Random Forest, Gradient Boosting (sklearn GBDT), AdaBoost, XGBoost, MLP, Other neural network (specify), Other (specify)

k-NN: straightforward baseline, suitable for multiclass classification problems. **Random Forest**: a common approach to this particular problem, several papers report good performance.

Example: "DummyClassifier: baseline sanity check.
Logistic Regression: interpretable linear baseline; works well with scaled numeric features.
Random Forest: captures non-linear interactions; strong tabular default."

6. Hyperparameters

This section asks you to identify your pipelines' hyperparameters and how you will tune them.

34. **Hyperparameters:** For each model *and preprocessing* step in your pipeline, read through its documentation (e.g., from sklearn's documentation) to identify available hyperparameters. Choose which hyperparameters you will tune and list them below. Give a justification for *why* you chose these hyperparameters. If this is a model we've covered in class, use your knowledge of the model. Otherwise, look for examples, documentation, or papers that suggest commonly tuned hyperparameters, and link these sources in your justification. If you are unsure which hyperparameters are most important, explain that.

k-NN:
- n_neighbors
- weights

Justification: n_neighbors is the main parameter that affects the complexity of the model, and using distance weighting may help the model account for noisy data.

Random Forest:
- n_estimators
- min_samples_leaf

Justification: Since random forest is an ensemble method, the number of trees is the primary determinant of complexity; ccp_alpha is comparatively less important. Including n_estimators and min_samples_leaf will control model complexity and help prevent overfitting.

Example:
Decision tree tuned hyperparameters:
- ccp_alpha
- max_depth

Justification: Most decision tree hyperparameters control complexity for pre- or post-pruning, and are therefore a bit redundant: we include one for each. The criterion HP is not very impactful, so we will not tune it.

35. **Initial Search Grid:** For each model/hyperparameter, specify the initial grid/distribution of hyperparameter values that will be used. If you are unsure, use documentation, examples, or papers to identify commonly used values.
    Tip: Before you do this step, consider calculating the complexity of tuning all of your hyperparameters, and consider removing some from automated tuning.

n_neighbors: discrete; distribution; 2-50; we'll consider higher values than normal since our dataset is large and noisy.
weights: discrete; grid; 'uniform', 'distance'; the only options available.
n_estimators: discrete; distribution; 80-160; the conference paper that our data came from used 120 for the full dataset.

Format: Tuned hyperparameter, parameter type (discrete/continuous), search space type, candidate values choice, explanation.

36. **Specify a hyperparameter tuning approach. Justify your answer.**
- Grid search
- Random search
- Other (describe)

We will be using an automated tuning method, probably either Optuna or Auto-sklearn.

37. **Specify a hyperparameter evaluation approach. Justify your answer.**
    - Holdout (specify train/validation split)
    - k-fold CV (specify k)
    - LOOCV
    - Other (describe)

Automated tuning may be more computationally intensive than doing it by hand, but a simple holdout approach will help reduce computation. Our dataset is large enough to allow for a separate validation set.

38. **Cost of Tuning:** Choose **one model/pipeline** with the greatest number of hyperparameters. Calculate the total number of settings being compared for that pipeline (preprocessing + model). Alternatively, if you have many (>10) hyperparameters, provide an upper bound for the number of settings. Please provide calculations.

k-NN: [2 weight values] * [49 neighbor values] = 98

Example (upper bound): [max # of HP values tested per HP] ^ [# of HPs]

39. **Select a strategy for reducing computation if needed** (e.g., start coarse w/ a small budget to eliminate some settings and adjust ranges, tune some less interactive hyperparameters independently first, and fix them).

If it turns out that the ranges are too large, we can start with a larger step size to narrow them down.

7. Algorithmic Bias

This section asks you to consider your model's impact on protected groups.

40. **Does this dataset include human subjects or demographic attributes that could relate to protected groups?**

- Yes
- No
- Not applicable

If “No” or “Not applicable”, skip to the next section.

41. **List demographic attributes you will treat as protected or fairness-relevant groups (if any).**

Attribute	Groups compared	Why fairness-relevant	Justification

Example:

- **Attribute:** Age, Sex/Gender, Race/Ethnicity, Other
- **Groups compared:** Female vs Male vs Non-binary/Other, 18–25 vs 26–40 vs 41+, Rural vs Urban
- **Why fairness-relevant:** protected group, access/resources, historical bias risk

42. **Which fairness metric(s) will you use?** Select one or more fairness metrics to evaluate bias across groups. **Why these metrics?** Briefly explain why the selected metrics match the intended use of the model.
    - Statistical Parity Difference (Selection Rate Difference)
    - Equal Opportunity Difference (TPR Difference)
    - TPR Difference (Recall Gap)
    - FPR Difference (False Positive Rate Gap)
    - Average Odds Difference (½(FPR diff + TPR diff))
    - Average Absolute Odds Difference/Average Odds Error (½(|FPR diff| + |TPR diff|))
    - Other

Justify your answer:

43. **Mitigating bias**: If you plan to mitigate detected bias, how do you plan to do so (this might be a novel aspect of your project)?

8. Execution Plan

This section asks you to describe how you will execute your project plan.

44. **Papers to read**: Describe 1 relevant paper you will read to help you with the technical aspects of your project (i.e., the prediction problem) and 1 paper or article to help you understand the context of the problem you are trying to solve (e.g., if detecting cancer from images, read an article on how doctors diagnose and treat cancer).

Technical aspects:
Mohammad Saiful Islam Mamun, Mohammad Ahmad Rathore, Arash Habibi Lashkari, Natalia Stakhanova, and Ali A. Ghorbani, "Detecting Malicious URLs Using Lexical Analysis", Network and System Security, Springer International Publishing, pp. 467-482, 2016.

Context:
https://zerolynx.com/en/blogs/news/tecnicas-avanzadas-ofuscacion-url

45. **Teammates and work division**: Each teammate should have clearly defined roles and aspects of the midterm milestone that they are responsible for.

Brian: Lead data selection and preprocessing.
Ryan: Lead model selection and implementation.
William: Lead Machine Learning hyperparameter tuning.

46. **Midterm milestone**: What will you complete by the Midway report? Experimental results of some kind are expected here.

By the midway report, we hope to have all of our data preprocessing completed. By that time, we will begin tuning hyperparameters and running the early stages of our model.

9. Classification/Regression

Complete at least one of the following sections:

- The classification section is applicable only if you are doing classification
- The regression section is applicable only if you are doing regression

When you are finished, export as a PDF (make sure to select **all tabs**) and submit on Gradescope.

Classification

9a. Classification

Complete only if you are solving a classification problem.

1. **What is the class distribution of your target variable?** Paste the value counts for each class of your target (include percentages if you have them). **Example**: Yes: 160 (16%)/No: 840 (84%)

Benign: 7781 (21%) Defacement: 7930 (22%) Malware: 6712 (18%) Phishing: 7587 (21%) Spam: 6697 (18%)

2. **Is there a minority class?** A minority class has noticeably fewer examples than the others.
   - Yes
   - No
   - Not sure

3. If yes: **Which class(es) are the minority class(es)?** (paste exact label(s))

N/A

4. **How will you encode the target attribute?**
   - LabelEncoder (specify which number maps to which class below)
   - OneHotEncoder
   - No encoding needed (explain the meaning of each number below)
   - Other (describe)

Benign = 0 **Spam = 1** **Phishing = 2** **Malware = 3** **Defacement = 4**

5. **Do you care more about correctly predicting one class correctly than the other?** (Binary classification only)
   - Yes, one class matters more
   - No, both classes are equally important
   - Not sure
   - Not applicable

If yes, which class matters more? (paste the exact label). Note: This is often the “positive” class (the outcome you’re most concerned about).

N/A

6. **What is the type of the target attribute?**
    - Binary
    - Ordinal
    - Nominal
7. **How balanced are the classes?**
    - Perfectly balanced
    - Some imbalance
    - Very large imbalance (e.g.,< 10% minority class)
8. **Does the class imbalance reflect the true real-world distribution or a data collection artifact?**
    - True real-world distribution
    - Likely data collection artifact
    - Not applicable
9. **Primary evaluation metrics.** Select the primary metric you will optimize (e.g., for hyperparameter tuning).
    - Accuracy
    - Precision
    - Recall
    - F1
    - ROC-AUC
    - PR-AUC
    - Other (specify)

Justification. Explain why this metric fits your target type, class balance, and goals.

The F1 score best fits our goals of developing an effective tuning capability from the predictive results of the model. We aspire to minimize false negatives so that generally malicious URLs are not classified as benign. Additionally, the various classes of malicious URLS are also desired to be classified appropriately since different resolution methods may be performed for each case (spam may just require being blocked while malware may require further analysis and eradication). These goals also, in turn, express the value of minimizing false positives, or the classification of benign URLs as potentially malicious, as this could interrupt the general operations of users. Since the F1 score balances these metrics of Precision (false positives) and Recall (false negatives), we select it as our primary evaluation metric.

10. **Select additional metrics you will report, even if they are not used for tuning.**
    - Accuracy

- Precision
- Recall
- F1
- ROC-AUC
- PR-AUC
- Other (specify)

Justification. Explain why this metric fits your target type, class balance, and goals.

Precision and Recall can be reported as a byproduct of the goal to utilize the F1 Score as the primary evaluation metric.

Accuracy is also a useful metric to report as it helps illustrate the overall effectiveness of our goal to properly predict whether URLs are benign or malicious and the various classes of malicious.

11. **What performance would seem "too good to be true" for your task? Briefly justify using one or more checks below.**

A model outcome that displayed 98-100% F1 Score would be considered as "too good to be true" performance. This suspicious result could be justified with a leakage check, as this almost perfect result suggests the split between training and test data was compromised.

Example:

- **Baseline (required):** Compare to a simple baseline (classification: majority-class accuracy; regression: predict mean/median).
- **External reference (if available):** If this dataset is from Kaggle or literature, report what others achieved and compare.
- **Leakage check:** Near-perfect performance may suggest leakage (target proxy or a bad split with overlap).

12. **What performance would mean you can definitely do better?** (Define an unacceptable baseline or minimum target.)

Currently, we would expect a model outcome that displayed 60% F1 Score or under to mean we could create a better model, as it does not provide much confidence in any effective usage of the final model to improve cybersecurity posture in use.